

Physics Notes

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Chapter 1

Mathematical Tools

1.1 Vector Calculus

For a scalar field ϕ and a vector field $\mathbf{A}$, the most common differential operators are

$$\nabla\phi, \quad \nabla \cdot \mathbf{A}, \quad \nabla \times \mathbf{A}, \quad \nabla^2\phi. \quad (1.1)$$

Theorem 1.1 (Divergence theorem). *If the boundary of a volume V is the closed surface ∂V , then*

$$\iiint_V (\nabla \cdot \mathbf{A}) \, dV = \oint_{\partial V} \mathbf{A} \cdot d\mathbf{S}. \quad (1.2)$$

Section Summary

Use this box to record key equations, assumptions, dimensional checks, and connections to other chapters.

Chapter 2

Classical Mechanics

2.1 Newton's Laws

Law 2.1 (Newton's second law). *The net force acting on a particle equals the time derivative of its momentum:*

$$\mathbf{F} = \frac{d\mathbf{p}}{dt}. \quad (2.1)$$

For constant mass, [equation \(2.1\)](#) becomes

$$\mathbf{F} = m\mathbf{a}. \quad (2.2)$$

2.2 Dimensional Analysis

The SI unit of force is $\text{N} = \text{kg m/s}^2$.

Common Pitfall

State the reference frame before writing the equations of motion. The standard form of Newton's second law applies directly only in an inertial frame.

Chapter 3

Electromagnetism

3.1 Maxwell's Equations

In vacuum, the differential form is

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}, \quad \nabla \cdot \mathbf{B} = 0, \quad (3.1)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (3.2)$$

3.2 A Simple Circuit Diagram

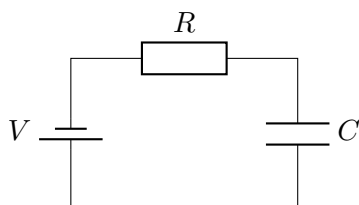


Figure 3.1: An example of a series RC circuit.

Chapter 4

Quantum Mechanics

4.1 States and Operators

A state vector is denoted by $|\psi\rangle$. In the position representation,

$$\psi(x, t) = \langle x | \psi(t) \rangle. \quad (4.1)$$

Position and momentum obey the canonical commutation relation

$$[\hat{x}, \hat{p}] = i\hbar. \quad (4.2)$$

The time-dependent Schrödinger equation is

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle. \quad (4.3)$$

For an early landmark in special relativity, see [1]. A standard modern introduction to quantum mechanics is [2].

Appendix A

Common Physical Constants

Table A.1: Common physical constants. Verify the latest CODATA values before using these data in precision work.

Quantity	Symbol	Value
Speed of light in vacuum	c	$2.997\,924\,58 \times 10^8 \text{ m/s}$
Elementary charge	e	$1.602\,176\,634 \times 10^{-19} \text{ C}$
Planck constant	h	$6.626\,070\,15 \times 10^{-34} \text{ J s}$

References

- [1] A. Einstein. “Zur Elektrodynamik bewegter Körper”. In: *Annalen der Physik* 322.10 (1905), pp. 891–921. DOI: [10.1002/andp.19053221004](https://doi.org/10.1002/andp.19053221004).
- [2] D. J. Griffiths and D. F. Schroeter. *Introduction to Quantum Mechanics*. 3rd ed. Cambridge: Cambridge University Press, 2018. DOI: [10.1017/9781316995433](https://doi.org/10.1017/9781316995433).